

AA3 – NFAs

(soluções)

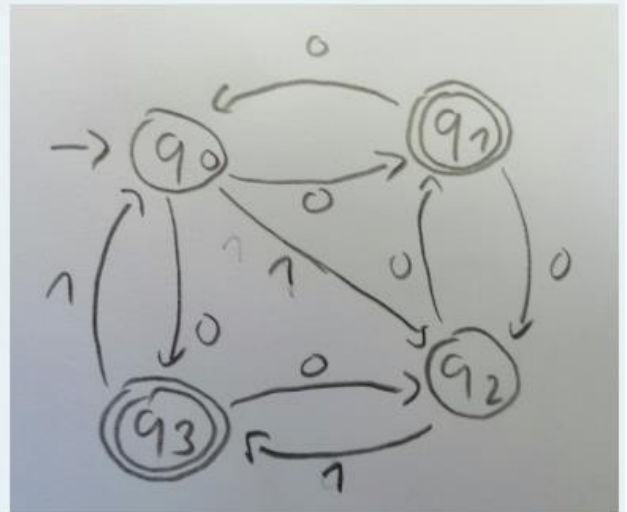
1

When using subset construction to build a DFA from a NFA, which states of the DFA become accepting?

(1/1 Ponto)

- ☒ Those states whose set contains at least one accepting state of the NFA.
- ☐ Those states whose set only contains accepting states of the NFA.
- ☐ Those states which have no transitions to other states.

2

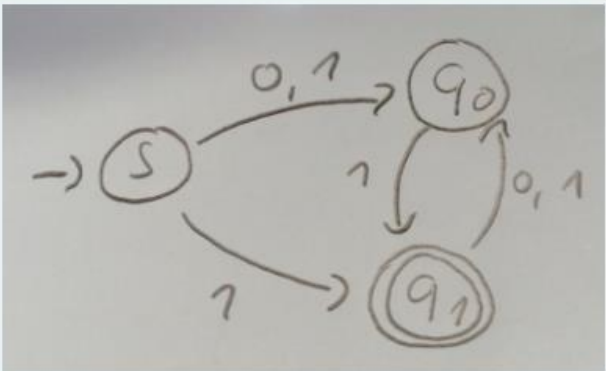


Considering the following NFA, if we build a corresponding DFA using the subset construction technique, which group has all its strings accepted by the DFA? 

(1/1 Ponto)

- ☐ Group1 composed of strings "0110", "0100", "0101" and "1010".
- ☒ Group 2 composed of strings "0110", "1000", "1101" and "1110".
- ☐ Group 3 composed of strings "1000", "1010", "1011" and "1100".

3

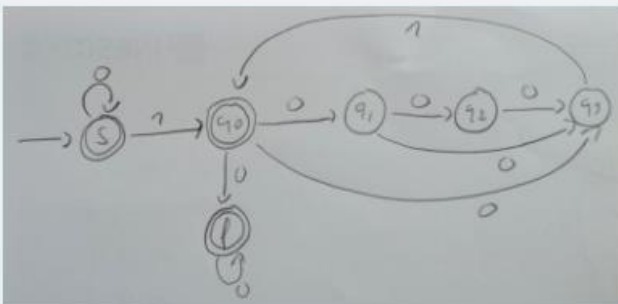


Considering the following NFA, if we build a corresponding DFA using the subset construction technique, how many states have a direct transition to the dead state, in the corresponding DFA? 

(1/1 Ponto)

- ☒ 1
☐ 2
☐ 3
☐ 4

4

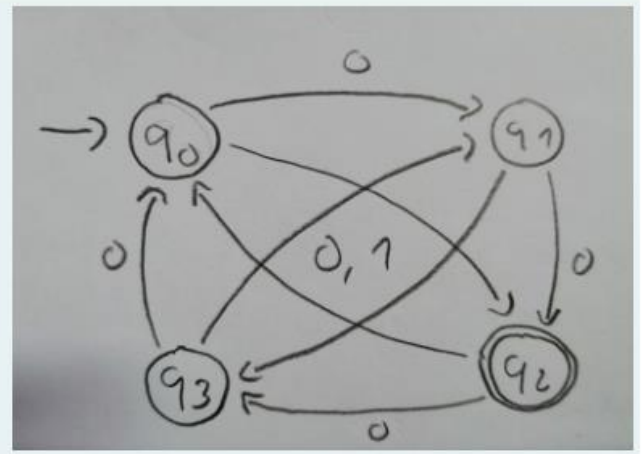


Considering the following NFA, which of the following options best describes the language accepted by it? 

(0/2 Pontos)

- ☐ Language composed of 0's and 1's, in which there can be no two 1's in a row.
- ☐ Language composed of 0's and 1's, in which there exists two 1's separated by one, two or three 0's.
- ☒ Language composed of 0's and 1's, in which every 1 is separated from another possible 1 by one, two or three 0's.

5



Considering the following NFA, if we build a corresponding DFA using the subset construction technique, how many states does the DFA has? **Solution: 7**
(2/2 Pontos)

6

Considering the NFA from the previous question, what is the result of $\delta^*(q_0, 01101)$?
(0/1 Ponto)

- ☐ {q0, q2}
- ☒ {q0, q1, q2}
- ☐ {q0, q2, q3}
- ☐ {q0, q1, q2, q3}